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Broadcasting a Host-Controlled Parameter to Circuitry Distributed at a Local-Bank Level for Usage-Based-Disturbance Mitigation

Abstract

Apparatuses and techniques for broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation are described. In an example aspect, a memory device includes usage-based-disturbance circuitry distributed at a local-bank level and bus circuitry. The bus circuitry broadcasts information provided by a memory controller to the usage-based-disturbance circuitry. The bus circuitry provides a means of efficient communication between the mode controller and the usage-based-disturbance circuitry to enable the memory controller to control some aspect of the usage-based-disturbance mitigation operation performed by the usage-based-disturbance circuitry. This means of communication enables the memory controller to dynamically adjust an operation and/or performance of the memory device with respect to usage-based-disturbance mitigation. Some memory devices may already include the bus circuitry, which enables these techniques to be implemented without significantly increasing signal routing complexity, die size, or cost.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/561,483, filed on Mar. 5, 2024, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND
[0002] Computers, smartphones, and other electronic devices rely on processors and memories. A processor executes code based on data to run applications and provide features to a user. The processor obtains the code and the data from a memory. The memory in an electronic device can include volatile memory (e.g., random-access memory (RAM)) and non-volatile memory (e.g., flash memory). Like the capabilities of a processor, the capabilities of a memory can impact the performance of an electronic device. This performance impact can increase as processors are developed that execute code faster and as applications operate on increasingly larger data sets that require ever-larger memories.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] Apparatuses of and techniques for broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation are described with reference to the following drawings. The same numbers are used throughout the drawings to reference like features and components:
[0004] FIG. **1** illustrates example apparatuses that can implement aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation;
[0005] FIG. **2** illustrates an example computing system that can implement aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation within a memory device;
[0006] FIG. **3** illustrates example data stored within rows of a memory array;
[0007] FIG. **4** illustrates an example memory device in which aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation may be implemented;
[0008] FIG. **5** illustrates an example arrangement of usage-based-disturbance circuitry on a die;
[0009] FIG. **6** illustrates an example architecture for broadcasting information provided at a global level to a local-bank level;
[0010] FIG. **7** illustrates an example transaction diagram for broadcasting information between a global level and a local-bank level of a memory device;
[0011] FIG. **8** illustrates a first example implementation of parameter update circuitry coupled to a bank-specific usage-based-disturbance circuit;

[0012] FIG. **9** illustrates a second example implementation of parameter update circuitry coupled to a bank-specific usage-based-disturbance circuit;
[0013] FIG. **10** illustrates a first example method of a memory device broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation; and
[0014] FIG. **11** illustrates a second example method of a memory device broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation.

DETAILED DESCRIPTION

Overview

[0015] Processors and memory work in tandem to provide features to users of computers and other electronic devices. As processors and memory operate more quickly together in a complementary manner, an electronic device can provide enhanced features, such as high-resolution graphics and artificial intelligence (AI) analysis. Some applications, such as those for financial services, medical devices, and advanced driver assistance systems (ADAS), can also demand more-reliable memories. These applications use increasingly reliable memories to limit errors in financial transactions, medical decisions, and object identification. However, in some implementations, more-reliable memories can sacrifice bit densities, power efficiency, and simplicity.
[0016] To meet the demands for physically smaller memories, memory devices can be designed with higher chip densities. Increasing chip density, however, can increase the electromagnetic coupling (e.g., capacitive coupling) between adjacent or proximate rows of memory cells due, at least in part, to a shrinking distance between these rows. With this undesired coupling, activation (or charging) of a first row of memory cells can sometimes negatively impact a second nearby row of memory cells. In particular, activation of the first row can generate interference, or crosstalk, that causes the second row to experience a voltage fluctuation. In some instances, this voltage fluctuation can cause a state (or value) of a memory cell in the second row to be incorrectly determined by a sense amplifier. Consider an example in which a state of a memory cell in the second row is a “1”. In this example, the voltage fluctuation can cause a sense amplifier to incorrectly determine the state of the memory cell to be a “0” instead of a “1”. Left unchecked, this interference can lead to memory errors or data loss within the memory device.
[0017] In some circumstances, a particular row of memory cells is activated repeatedly in an unintentional or intentional (sometimes malicious) manner. Consider, for instance, that memory cells in an Rth row are subjected to repeated activation, which causes one or more memory cells in an adjacent row (e.g., within an R+1 row, an R+2 row, an R−1 row, and/or an R−2 row) to change states. This effect is referred to as a usage-based disturbance. The occurrence of usage-based disturbance can lead to the corruption or changing of contents within the affected row of memory.
[0018] Some memory devices perform usage-based-disturbance mitigation in an independent manner without input from a memory controller (or a host device). In this situation, the memory controller has zero control or influence regarding the usage-based-disturbance mitigation process. The memory controller, for instance, is unable to specify or configure aspects of the usage-based-disturbance mitigation process to achieve a desired level of performance. If the memory controller may communicate with or is designed to support operation with different memory devices from different manufactures, it can be challenging for the memory controller to take into account dissimilarities regarding operation and/performance of these memory devices regarding usage-based-disturbance mitigation.
[0019] It may therefore be desirable to provide a means by which the memory controller can control some aspect of the usage-based-disturbance mitigation operation performed by a memory device. In addition to enabling the memory controller to configure some aspect of the usage-based-disturbance mitigation operation, this control can further enable the memory controller to similarly configure or adjust usage-based-disturbance-mitigation performance of various devices from

different manufacturers. An architecture of the memory device, however, may not be conducive to enabling the memory controller to control an aspect of usage-based-disturbance mitigation, as further explained below.

[0020] An example memory device can have multiple instances of a circuit that performs usage-based-disturbance mitigation implemented at a local-bank level of the memory device. This architecture enables each circuit to mitigate usage-based disturbance within a particular bank (or subset of banks) that is proximate to the circuit. The multiple instances of these circuits and their distribution at the local-bank level, however, can make it challenging to route information provided by the memory controller to the individual circuits. It can be particularly challenging to implement a means of communication between the memory controller and the individual circuits without significantly increasing signal routing complexity, increasing die size, or increasing cost.

[0021] To address this and other issues regarding usage-based disturbance, this document describes aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation. In an example aspect, a memory device includes usage-based-disturbance circuitry distributed at a local-bank level and bus circuitry. The bus circuitry broadcasts information provided by a memory controller (or a host device) to the usage-based-disturbance circuitry. The bus circuitry provides a means of efficient communication between the mode controller and the usage-based-disturbance circuitry to enable the memory controller to control some aspect of the usage-based-disturbance mitigation operation performed by the usage-based-disturbance circuitry. This means of communication enables the memory controller to dynamically adjust an operation and/or performance of the memory device with respect to usage-based-disturbance mitigation.

[0022] In some memory devices, the bus circuitry can serve other purposes during other time intervals or during other modes of operation. For example, the bus circuitry can broadcast initialization information to the usage-based-disturbance circuitry and/or other circuitry implemented at the local-bank level as part of a process that initializes the memory device. Additionally or alternatively, the bus circuitry can broadcast information associated with a test to the other circuitry as part of a process that tests some aspect of the memory device. By utilizing and/or repurposing the bus circuitry to support broadcasting of a host-controlled parameter to the usage-based-disturbance circuitry during a normal operational mode, the techniques described herein can be implemented without significantly increasing signal routing complexity, increasing die size, or increasing cost.

Example Operating Environments

[0023] FIG. **1** illustrates, at **100** generally, an example operating environment including an apparatus **102** that can implement broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation. The apparatus **102** can include various types of electronic devices, including an internet-of-things (IoT) device **102-1**, tablet device **102-2**, smartphone **102-3**, notebook computer **102-4**, passenger vehicle **102-5**, server computer **102-6**, and server cluster **102-7** that may be part of cloud computing infrastructure, a data center, or a portion thereof (e.g., a printed circuit board (PCB)). Other examples of the apparatus **102** include a wearable device (e.g., a smartwatch or intelligent glasses), entertainment device (e.g., a set-top box, video dongle, smart television, a gaming device), desktop computer, motherboard, server blade, consumer appliance, vehicle, drone, industrial equipment, security device, or sensor, or electronic components thereof. Each type of apparatus can include one or more components to provide computing functionalities or features.

[0024] In example implementations, the apparatus **102** can include at least one host device **104**, at least one interconnect **106**, and at least one memory device **108**. The host device **104** can include at least one processor **110**, at least one cache memory **112**, and a memory controller **114**. The memory device **108**, which can also be realized with a memory module, can include, for example, a dynamic random-access memory (DRAM) die or module (e.g., Low-Power Double Data Rate synchronous DRAM (LPDDR SDRAM)). The DRAM die or module can include a three-dimensional (3D) stacked DRAM device, which may be a high-bandwidth memory (HBM) device or a hybrid memory cube (HMC) device. The memory device **108** can operate as a main memory for the apparatus **102**. Although not illustrated, the apparatus **102** can also include storage memory. The storage memory can include, for example, a storage-class memory device (e.g., a flash memory, hard disk drive, solid-state drive, phase-change memory (PCM), or memory employing 3D XPoint™).

[0025] The processor **110** is operatively coupled to the cache memory **112**, which is operatively coupled to the memory controller **114**. The processor **110** is also coupled, directly or indirectly, to the memory controller **114**. The host device **104** may include other components to form, for instance, a system-on-a-chip (SoC). The processor **110** may include a general-purpose processor, central processing unit, graphics processing unit (GPU), neural network engine or accelerator, application-specific integrated circuit (ASIC), field-programmable gate array (FPGA) integrated circuit (IC), or communications processor (e.g., a modem or baseband processor).

[0026] In operation, the memory controller **114** can provide a high-level or logical interface between the processor **110** and at least one memory (e.g., an external memory). The memory controller **114** may be realized with any of a variety of suitable memory controllers (e.g., a double-data-rate (DDR) memory controller that can process requests for data stored on the memory device **108**). Although not shown, the host device **104** may include a physical interface (PHY) that transfers data between the memory controller **114** and the memory device **108** through the interconnect **106**. For example, the physical interface may be an interface that is compatible with a DDR PHY Interface (DFI) Group interface protocol. The memory controller **114** can, for example, receive memory requests from the processor **110** and provide the memory requests to external memory with appropriate formatting, timing, and reordering. The memory controller **114** can also forward to the processor **110** responses to memory requests received from external memory.

[0027] The host device **104** is operatively coupled, via the interconnect **106**, to the memory device **108**. In some examples, the memory device **108** is connected to the host device **104** via the interconnect **106** with an intervening buffer or cache. The memory device **108** may operatively couple to storage memory (not shown). The host device **104** can also be coupled, directly or indirectly via the interconnect **106**, to the memory device **108** and the storage memory. The interconnect **106** and other interconnects (not illustrated in FIG. **1**) can transfer data between two or more components of the apparatus **102**. Examples of the interconnect **106** include a bus (e.g., a unidirectional or bidirectional bus), switching fabric, or one or more wires that carry voltage or current signals. The interconnect **106** can propagate one or more communications **116** between the host device **104** and the memory device **108**. For example, the host device **104** may transmit a memory request to the memory device **108** over the interconnect **106**. Also, the memory device **108** may transmit a corresponding memory response to the host device **104** over the interconnect **106**.

[0028] The illustrated components of the apparatus **102** represent an example architecture with a hierarchical memory system. A hierarchical memory system may include memories at different levels, with each level having memory with a different speed or capacity. As illustrated, the cache memory **112** logically couples the processor **110** to the memory device **108**. In the illustrated implementation, the cache memory **112** is at a higher level than the memory device **108**. A storage memory, in turn, can be at a lower level than the main memory (e.g., the memory device **108**). Memory at lower hierarchical levels may have a decreased speed but increased capacity relative to memory at higher hierarchical levels.

[0029] The apparatus **102** can be implemented in various manners with more, fewer, or different components. For example, the host device **104** may include multiple cache memories (e.g., including multiple levels of cache memory) or no cache memory. In other implementations, the host device **104** may omit the processor **110** or the memory controller **114**. A memory (e.g., the memory device **108**) may have an “internal” or “local” cache memory. As another example, the apparatus **102** may include cache memory between the interconnect **106** and the memory device **108**. Computer engineers can also include any of the illustrated components in distributed or shared memory systems.

[0030] Computer engineers may implement the host device **104** and the various memories in multiple manners. In some cases, the host device **104** and the memory device **108** can be disposed on, or physically supported by, a printed circuit board (e.g., a rigid or flexible motherboard). The host device **104** and the memory device **108** may additionally be integrated together on an integrated circuit or fabricated on separate integrated circuits and packaged together. The memory device **108** may also be coupled to multiple host devices **104** via one or more interconnects **106** and may respond to memory requests from two or more host devices **104**. Each host device **104** may include a respective memory controller **114**, or the multiple host devices **104** may share a memory controller **114**. This document describes with reference to FIG. **1** an example computing system architecture having at least one host device **104** coupled to a memory device **108**.

[0031] Two or more memory components (e.g., modules, dies, banks, or bank groups) can share the electrical paths or couplings of the interconnect **106**. The interconnect **106** can include at least one command-and-address bus (CA bus) and at least one data bus (DQ bus). The command-and-address bus can transmit addresses and commands from the memory controller **114** of the host device **104** to the memory device **108**, which may exclude propagation of data. The data bus can propagate data between the memory controller **114** and the memory device **108**. The memory device **108** may also be implemented as any suitable memory including, but not limited to, DRAM, SDRAM, three-dimensional (3D) stacked DRAM, DDR memory, or LPDDR memory (e.g., LPDDR DRAM or LPDDR SDRAM).

[0032] The memory device **108** can form at least part of the main memory of the apparatus **102**. The memory device **108** may, however, form at least part of a cache memory, a storage memory, or a system-on-chip of the apparatus **102**. The memory device **108** includes usage-based-disturbance (UBD) circuitry **118** (UBD **118**) and parameter update circuitry **120**. The usage-based-disturbance circuitry **118** and the parameter update circuitry **120** can be implemented using combinations of hardware, fixed circuit circuitry, software, and/or firmware.

[0033] The usage-based-disturbance (UBD) circuitry **118** is implemented at a local-bank level **122**. This means that different components of the usage-based-disturbance circuitry **118** are coupled to different banks (or different sets of banks) within a bank group. In some implementations, components implemented at the local-bank level **122** are disposed in close proximity to the corresponding bank (or banks) to simplify and/or reduce signal routing.

[0034] In contrast, the parameter update circuitry **120** is implemented at a global level **124** (or a central level). This means that components of the parameter update circuitry **120** are coupled to other components that are implemented at the local-bank level **122**. Accordingly, the components of the parameter update circuitry **120** can be considered to be coupled (or indirectly coupled) to multiple banks. In some cases, the multiple banks can include all of the banks within a bank group. In some implementations, a component implemented at the global level **124** may be disposed relatively centrally to the multiple banks to more evenly distribute the routing between a component at the global level **124** and multiple components at the local-bank level **122**.

[0035] Generally speaking, the usage-based-disturbance circuitry **118** mitigates usage-based disturbance within the memory device **108**. The usage-based-disturbance circuitry **118** includes a plurality of bank-specific usage-based-disturbance circuits **126-1** to **126**-N (bank-specific UBD circuits **126-1** to **126**-N), where N is a positive integer. The bank-specific usage-based-disturbance circuits **126-1** to **126**-N mitigate usage-based disturbance mitigation within its corresponding bank (or banks). The bank-specific usage-based-disturbance circuits **126-1** to **126**-N represent components of the usage-based-disturbance circuitry **118** that are distributed at the local-bank level **122**.

[0036] Each bank-specific usage-based-disturbance circuit **126** can include at least one counter circuit for tracking row activations, at least one comparator circuit for detecting conditions associated with usage-based disturbance, at least one queue for managing refresh operations responsive to a usage-based disturbance, and/or at least one error-correction-code (ECC) circuit for detecting and/or correcting bit errors associated with usage-based disturbance. One aspect of usage-based disturbance mitigation involves keeping track of how often a row is activated or accessed since a last refresh. In particular, the bank-specific usage-based-disturbance circuit **126** performs an array counter update procedure using the counter circuit to update an activation count associated with an activated row. During the array counter update procedure, the bank-specific usage-based-disturbance circuit **126** reads the activation count that is stored within the activated row, increments the activation count, and writes the updated activation count to the activated row. By maintaining the activation count, the bank-specific usage-based-disturbance circuit **126** can determine when to perform a refresh operation to reduce the risk of usage-based disturbance. For example, when the comparator circuit determines that the activation count meets or exceeds a threshold (e.g., a mitigation threshold), the bank-specific usage-based-disturbance circuit **126** can perform a procedure to refresh one or more rows that are near the activated row to mitigate the usage-based disturbance.

[0037] The parameter update circuitry **120** implements, at least in part, the broadcasting of a host-controlled parameter to circuitry distributed at the local-bank level **122** for usage-based-disturbance mitigation. The parameter update circuitry **120** includes bus circuitry **128**, at least one mode register **130**, and at least one auxiliary memory **132**. The bus circuitry **128** is implemented at the global level **124** and broadcasts (e.g., passes) information provided at the global level **124** to the local-bank level **122**. For usage-based-disturbance mitigation purposes, the bus circuitry **128** broadcasts information provided by the memory controller **114** (e.g., the host device **104**) to the usage-based-disturbance circuitry **118**. The bus circuitry **128** includes at least one communication bus and logic circuitry for formatting information that is to be communicated via the communication bus.

[0038] In some implementations, the bus circuitry **128** represents a new component that is integrated within the memory device **108** to broadcast information to the usage-based-disturbance circuitry **118**. In other implementations, the memory device **108** already includes the bus circuitry **128**, which is used for broadcasting other information to other components of the memory device **108**. In some cases, this bus circuitry **128** is not used during a normal operational mode (e.g., during a time period in which the memory device **108** can receive a mode register write command). Sometimes the bus circuitry **128** may be utilized during initialization and/or off-line testing of the memory device **108**. As such, capabilities of the bus circuitry **128** can be expanded to support the broadcasting of a host-controlled parameter to circuitry distributed at the local-bank level **122** for usage-based-disturbance mitigation. In general, any bus circuitry **128** within the memory device **108** can be operated in a manner that supports the broadcasting of the host-controlled parameter to circuitry distributed at the local-bank level **122** for usage-based-disturbance mitigation so long as the broadcasting of the host-controlled parameter does not conflict or interfere with other operations of the memory device **108** that utilize the bus circuitry **128**.

[0039] The mode register **130** is also implemented at the global level **124** and stores information (e.g., at least one host-controlled parameter) that is provided by the memory controller **114**. In some implementations, the mode register **130** (or multiple mode registers **130**) store multiple types of information (e.g., different host-controlled parameters). The stored information enables the memory controller **114** to control at least one aspect of an operation performed by the usage-based-disturbance circuitry **118** for mitigating usage-based disturbance. In some cases, the information stored by the mode register **130** can be directly used by the usage-based-disturbance circuitry **118** to perform an operation associated with usage-based-disturbance mitigation. For example, the information can determine a value of a parameter (e.g., a mitigation threshold or an alert threshold) used in the operation. In this case, the parameter update circuitry **120** can directly provide the information to the bus circuitry **128** for broadcasting.

[0040] In other cases, the information stored by the mode register **130** can be indirectly used by the usage-based-disturbance circuitry **118** to perform the operation associated with usage-based-disturbance mitigation. For example, the information can represent an offset relative to a value of a default parameter. In this case, the parameter update circuitry **120** can further process the information and provide the resulting processed information to the bus circuitry **128** for broadcasting to the usage-based-disturbance circuitry **118**. Aspects of broadcasting a host-controlled parameter to circuitry distributed at the local-bank level **122** for usage-based-disturbance mitigation can be readily detected based on a mode register write command associated with the mode register **130** causing bus circuitry **128** to broadcast information stored by the mode register **130**.

[0041] The auxiliary memory **132** is implemented at the global level **124** and includes a secondary memory that is different than the memory associated with read and write commands sent by the memory controller **114**. In some implementations, the auxiliary memory **132** is a non-volatile memory such as a fuse array, a flash memory, metal bits, a programmable read-only memory, a one-time programmable memory, and so on. Other implementations are also possible in which the auxiliary memory **132** is a volatile memory, such as a cache memory, a random-access memory (RAM), or a portion of a memory array of the memory device **108** that is designated for auxiliary purposes.

[0042] The auxiliary memory **132** stores information that can be broadcast to the local-bank level **122** using the bus circuitry **128**. In some implementations, the information stored by the auxiliary memory **132** includes a default parameter associated with the information stored by the mode register **130**. Additionally, the information stored by the auxiliary memory **132** can include information used during another mode of operation that differs from the normal operational mode. Example types of information can include information used for testing some aspect of the memory device **108** and applied during a test mode and/or information used to initialize some aspect of the memory device **108** and applied during an initialization mode. The mode register **130** and the auxiliary memory **132** represent non-transitory media. A computing system that includes the usage-based-disturbance circuitry **118** and the parameter update circuitry **120** is further described with respect to FIG. **2**.

[0043] FIG. **2** illustrates an example computing system **200** that can implement aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation. In some implementations, the computing system **200** includes at least one memory device **108**, at least one interconnect **106**, and at least one processor **202**. The memory device **108** can include, or be associated with, at least one memory array **204**, at least one interface **206**, and control circuitry **208** (or periphery circuitry) operatively coupled to the memory array **204**. The memory array **204** can include an array of memory cells, including but not limited to memory cells of DRAM, SDRAM, three-dimensional (3D) stacked DRAM, DDR memory, LPDDR SDRAM, and so forth. The memory array **204** and the control circuitry **208** may be components on a single semiconductor die or on separate semiconductor dies. The memory array **204** or the control circuitry **208** may also be distributed across multiple dies. This control circuitry **208** may manage traffic on a bus that is separate from the interconnect **106**.

[0044] The control circuitry **208** can include various components that the memory device **108** can use to perform various operations. These operations can include communicating with other devices, managing memory performance, performing refresh operations (e.g., self-refresh operations or auto-refresh operations), and performing memory read or write operations. For example, the control circuitry **208** can include at least one instance of array control logic **210**, clock circuitry **212**, the usage-based-disturbance circuitry **118**, and the parameter update circuitry **120**. The array control logic **210** can include circuitry that provides command decoding, address decoding, input/output functions, amplification circuitry, power supply management, power control modes, and other functions. The clock circuitry **212** can synchronize various memory components with one or more external clock signals provided over the interconnect **106**, including a command-and-address clock or a data clock. The clock circuitry **212** can also use an internal clock signal to synchronize memory components and may provide timer functionality

[0045] In one aspect, the usage-based-disturbance circuitry **118** can represent another part of the control circuitry **208**. The usage-based-disturbance circuitry **118** can be coupled to a set of memory cells within the memory array **204** that store usage-based-disturbance data **216**. The usage-based-disturbance data **216** can include information such as an activation count, which represents a quantity of times one or more rows within the memory array **204** have been activated (or accessed) by the memory device **108**. In example implementations, each row of the memory array **204** includes a subset of memory cells that stores the usage-based-disturbance data **216** associated with that row.

[0046] The interface **206** can couple the control circuitry **208** or the memory array **204** directly or indirectly to the interconnect **106**. In some implementations, the usage-based-disturbance circuitry **118**, the parameter update circuitry **120**, the array control logic **210**, and the clock circuitry **212** can be part of a single component (e.g., the control circuitry **208**). In other implementations, one or more of the usage-based-disturbance circuitry **118**, the parameter update circuitry **120**, the array control logic **210**, or the clock circuitry **212** may be implemented as separate components, which can be provided on a single semiconductor die or disposed across multiple semiconductor dies. These components may individually or jointly couple to the interconnect **106** via the interface **206**.

[0047] The interconnect **106** may use one or more of a variety of interconnects that communicatively couple together various components and enable commands, addresses, or other information and data to be transferred between two or more components (e.g., between the memory device **108** and the processor **202**). Although the interconnect **106** is illustrated with a single line in FIG. **2**, the interconnect **106** may include at least one bus, at least one switching fabric, one or more wires or traces that carry voltage or current signals, at least one switch, one or more buffers, and so forth. Further, the interconnect **106** may be separated into at least a command-and-address bus and a data bus.

[0048] In some aspects, the memory device **108** may be a "separate" component relative to the host device **104** (of FIG. **1**) or any of the processors **202**. The separate components can include a printed circuit board, memory card, memory stick, and memory module (e.g., a single in-line memory module (SIMM) or dual in-line memory module (DIMM)). Thus, separate physical components may be located together within a same housing of an electronic device or may be distributed over a server rack, a data center, and so forth. Alternatively, the memory device **108** may be integrated with other physical components, including the host device **104** or the processor **202**, by being combined on a printed circuit board or in a single package or a system-on-chip.

[0049] As shown in FIG. **2**, the processors **202** may include a computer processor **202-1**, a baseband processor **202-2**, and an application processor **202-3**, coupled to the memory device **108** through the interconnect **106**. The processors **202** may include or form a part of a central processing unit, graphics processing unit, system-on-chip, application-specific integrated circuit, or field-programmable gate array. In some cases, a single processor can comprise multiple processing resources, each dedicated to different functions (e.g., modem management, applications, graphics, central processing). In some implementations, the baseband processor **202-2** may include or be coupled to a modem (not illustrated in FIG. **2**) and referred to as a modem processor. The modem or the baseband processor **202-2** may be coupled wirelessly to a network via, for example, cellular, Wi-Fi®, Bluetooth®, near field, or another technology or protocol for wireless communication.

[0050] In some implementations, the processors **202** may be connected directly to the memory device **108** (e.g., via the interconnect **106**). In other implementations, one or more of the processors **202** may be indirectly connected to the memory device **108** (e.g., over a network connection or through one or more other devices). The memory array **204** is further described with respect to

FIG. **3**.

[0051] FIG. **3** illustrates example data stored within rows of the memory array **204**. The memory array **204** includes multiple rows **302** of memory cells. For example, the memory array **204** depicted in FIG. **3** includes rows **302-1**, **302-2** . . . **302**-R, where R represents a positive integer. Each row **302** is associated with an address **304** (e.g., a row address, a memory row address, or a memory address). For example, the first row **302-1** has a first address **304-1**, the second row **302-2** has a second address **304-2**, and an Rth row **302**-R has an Rth address **304**-R.

[0052] Each of the rows **302** can store normal data **306** within a first subset of the memory cells associated with that row **302**. The normal data **306** represents data that is read from or written to the memory device **108** during normal memory operations (e.g., during normal read or write operations). The normal data **306**, for example, can include data that is transmitted by the memory controller **114** and is written to one or more rows **302** of the memory array **204**.

[0053] In addition to the normal data **306**, each of the rows **302** can store usage-based-disturbance data **216** within a second subset of the memory cells associated with that row **302**. The usage-based-disturbance data **216** includes information that enables the usage-based-disturbance circuitry **118** to mitigate usage-based disturbance. In an example implementation, the usage-based-disturbance data **216** includes an activation count **308**.

[0054] In this example, the first row **302-1** stores first normal data **306-1** within a first subset of memory cells of the first row **302-1** and stores first usage-based-disturbance data **216-1** within a second subset of the memory cells of the first row **302-1**. The first usage-based-disturbance data **216-1** includes a first activation count **308-1**, which represents a quantity of times the first row **302-1** has been activated since a last refresh. As another example, the second row **302-2** stores second normal data **306-2** within a first subset of memory cells within the second row **302-2** and stores second usage-based-disturbance data **216-2** within a second subset of the memory cells within the second row **302-2**. The second usage-based-disturbance data **216-2** includes a second activation count **308-2**, which represents a quantity of times the second row **302-2** has been activated since a last refresh. Additionally, the Rth row **302**-R stores Rth normal data **306**-R within a first subset of memory cells within the Rth row **302**-R and stores Rth usage-based-disturbance data **216**-R within a second subset of the memory cells within the Rth row **302**-R. The Rth usage-based-disturbance data **216**-R includes an Rth activation count **308**-R, which represents a quantity of times the Rth row **302**-R has been activated since a last refresh.

[0055] The usage-based-disturbance data **216** also includes information or is formatted (e.g., coded) in such a way as to support error detection. In this example, the usage-based disturbance data **218** includes at least one parity bit **310** to enable detection of a faulty activation count **308** using a parity check. For instance, the usage-based-disturbance data **216-1**, **216-2**, and **216**-R respectively include parity bits **310-1**, **310-2**, and **310**-R. Other implementations are also possible in which the usage-based-disturbance data **216** is coded in a manner that supports any of the error detection tests described above, such as the error-correcting-code check.

Example Techniques and Hardware

[0056] FIG. **4** illustrates an example memory device **108** in which aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation can be implemented. The memory device **108** includes a memory module **402**, which can include multiple dies **404**. As illustrated, the memory module **402** includes a first die **404-1**, a second die **404-2**, a third die **404-3**, and a Dth die **404**-D, with D representing a positive integer. One or more of the dies **404-1** to **404**-D can include the usage-based-disturbance circuitry **118** and the parameter update circuitry **120**. The memory module **402** can be a SIMM or a DIMM. As another example, the memory module **402** can interface with other components via a bus interconnect (e.g., a Peripheral Component Interconnect Express (PCIe®) bus). The memory device **108** illustrated in FIGS. **1** and **2** can correspond, for example, to multiple dies (or dice) **404-1** through **404**-D, or a memory module **402** with two or more dies **404**. As shown, the memory module **402** can include one or more electrical contacts **406** (e.g., pins) to interface the memory module **402** to other components.

[0057] The memory module **402** can be implemented in various manners. For example, the memory module **402** may include a printed circuit board, and the multiple dies **404-1** through **404**-D may be mounted or otherwise attached to the printed circuit board. The dies **404** (e.g., memory dies) may be arranged in a line or along two or more dimensions (e.g., forming a grid or array). The dies **404** may have a similar size or may have different sizes. Each die **404** may be similar to another die **404** or different in size, shape, data capacity, or control circuitries. The dies **404** may also be positioned on a single side or on multiple sides of the memory module **402**.

[0058] One or more of the dies **404-1** to **404**-D include the usage-based-disturbance circuitry **118**, the parameter update circuitry **120**, and bank groups **408-1** to **408**-G, with G representing a positive integer. Each bank group **408** includes at least two banks **410**, such as banks **410-1** to **410**-B, with B representing a positive integer. In some implementations, the die **404** includes a plurality of bank-specific usage-based-disturbance circuits **126**, such as the bank-specific usage-based-disturbance circuits **126-1** to **126**-N, which mitigate usage-based disturbance across the banks **410** of the bank group **408**. For example, the bank-specific usage-based-disturbance circuitries (e.g., **126-1** to **126**-B) can respectively mitigate usage-based disturbance for respective banks **410** (e.g., banks **410-1** to **410**-B) of one bank group **408**, where B is less than N. In this case, each bank-specific usage-based-disturbance circuit **126** mitigates usage-based disturbance for a single bank **410** within one of the bank groups **408**. In another example, one or more of the bank-specific usage-based-disturbance circuits **126** mitigate usage-based disturbance for a subset of the banks **410** associated with one of the bank groups **408**. In this case, the subset of the banks **410** includes at least two banks **410**. The relationship between the banks **410** and bank-specific usage-based-disturbance circuits **126** are further described with respect to FIG. **5**.

[0059] FIG. **5** illustrates an example arrangement of multiple bank-specific usage-based-disturbance circuits **126** and the parameter update circuitry **120** on a die **404**. The die **404** includes bank-specific circuitry **502** and bank-shared circuitry **504**. The bank-specific circuitry **502** includes components that are implemented at the local-bank level **122** and are associated with a particular bank **410** or a subset of banks **410** (e.g., a proper subset of banks **410**). For example, the bank-specific circuitry **502** includes the banks **410-1**, **410-2** . . . **410**-(B/2), **410**-(B/2+1), **410**-(B/2+2) . . . **410**-B and the bank-specific usage-based-disturbance circuits **126-1**, **126-2** . . . **126**-(B/2), **126**-(B/2+1), **126**-(B/2+2) . . . **126**-B. The bank-specific usage-based-disturbance circuits **126-1** to **126**-B are respectively coupled to the banks **410-1** to **410**-B. In some cases, subsets of the banks **410-1** to **410**-B are associated with different bank groups **408**. In an example implementation, the die **404** includes 32 banks **410** (e.g., B equals 32). The 32 banks **410** form eight bank groups **408** (e.g., G equals 8), with each bank group **408** including four of the banks **410**. In other cases, the banks **410-1** to **410**-B are associated with a single bank group **408**.

[0060] The bank-shared circuitry **504** includes components that are implemented at the global level **124** and are associated with multiple banks **410**. These components perform operations associated with the multiple banks **410**. Example components of the bank-shared circuitry **504** include the parameter update circuitry **120**, which can broadcast information to the multiple bank-specific usage-based-disturbance circuits **126-1** to **126**-B.

[0061] On the die **404**, the bank-specific circuitry **502** may be positioned on two opposite sides of the bank-shared circuitry **504**. Explained another way, the bank-shared circuitry **504** can be centrally positioned on the die **404**. As such, the parameter update circuitry **120** can be positioned closer to a center of the die **404** compared to edges of the die **404**. Positioning the bank-shared circuitry **504** in the center enables routing between the bank-shared circuitry **504** and the bank-specific circuitry **502** to be simplified.

[0062] Consider a first axis **506** (e.g., x axis **505**) and a second axis **508** (e.g., y axis **510**), which is perpendicular to the first axis **506** In FIG. **5**, the first axis **506** is depicted as a “horizontal” axis, and the second axis **510** is depicted as a "vertical" axis. Components of the bank-shared circuitry **504** are distributed across the second axis **510**. A first set of the banks (e.g., banks **410-1** to bank **410**-(B/2)) are arranged along the second axis **510** on a "left" side of the bank-shared circuitry **504**, and a second set of the banks (e.g., banks **410**-(B/2+1) to **410**-B)) are arranged along the second axis **510** on a "right" side of the bank-shared circuitry **504**. The bank-specific usage-based-disturbance circuits **126-1** to **126**-B are positioned between the corresponding banks **410-1** to **410**-B and the bank-shared circuitry **504**. By positioning the parameter update circuitry **120** in a central location between the bank-specific usage-based-disturbance circuits **126-1** to **126**-B, it can be easier to route signals between the parameter update circuitry **120** and the bank-specific usage-based-disturbance circuits **126-1** to **126**-B. Operations of the bank-specific usage-based-disturbance circuits **126** are further described with respect to FIG. **6**.

[0063] FIG. **6** illustrates an example architecture for broadcasting information provided at the global level **124** to the local-bank level **122**. The architecture includes the usage-based-disturbance circuitry **118** and other circuitry **602**, which are implemented at the local-bank level **122**. The usage-based-disturbance circuitry **118** includes the plurality of bank-specific usage-based-disturbance circuits **126**, such as the bank-specific usage-based-disturbance circuits **126-1**, **126-2** . . . **126**-N.

[0064] The other circuitry **602** may include one or more circuits for performing operations other than usage-based-disturbance mitigation operations. Such operations may include test mode operations in which the memory device **108** may be subjected to various tests to ensure the functionality, reliability, and performance of the memory device **108**. Still other operations may include initializing components of the memory device **108** upon start-up. In this example, the other circuitry **602** includes a plurality of other circuits **604**, such as the other circuits **604-1**, **604-2** . . . **604**-Q, where Q represents a positive integer that may or may not be equal to N.

[0065] The architecture also includes the parameter update circuitry **120**, which is implemented at the global level **124**. The parameter update circuitry **120** includes the bus circuitry **128**, the mode register **130**, and the auxiliary memory **132**. In an example implementation, the mode register **130** facilitates control by and/or communication with the memory controller **114** (e.g., with the host device **104**). The mode register **130** enables the memory controller **114** to provide and/or modify at least one host-controlled parameter **606** that is referenced or utilized to perform a usage-based disturbance mitigation operation by the bank-specific usage-based-disturbance circuits **126**. In a sense, the host-controlled parameter **606** can be dynamically changed by the host device **104**. As such, the host-controlled parameter **606** can represent an "adaptive" parameter (e.g., a "dynamic" parameter or an "adjustable" parameter).

[0066] The auxiliary memory **132** can store one or more parameters that can be used to control an operation of the memory device **108**. In a scenario where the memory device **108** operates in a plurality of modes, the auxiliary memory **132** can include a plurality of memory sections configured to store a plurality of different parameters. For example, the auxiliary memory **132** can include a first memory section to store a default parameter **610** that represents a default value of the host-controlled parameter **606**. The default parameter **610** can be used to initialize the usage-based-disturbance circuitry **118**. The auxiliary memory **132** can also include a second memory section to store a test parameter **612** operable for testing the memory device **108**. Of course, these are not meant to be limiting, as one or more other parameters **614** operable for controlling activities of the memory device **108** can also be stored in the auxiliary memory **132** and passed to the other circuitry **602**. In some instances, the one or more other parameters **614** can be used to initialize the memory device **108** (or more specifically the other circuitry **602**).

[0067] The bank-specific usage-based-disturbance circuits **126-1** to **126**-N and the other circuits **604-1** to **604**-Q are each coupled to a bus **608** (e.g., a communication bus) of the bus circuitry **128**. The usage-based-disturbance circuitry **118** and the other circuitry **602** can both receive information (e.g., data or parameters) from the bus **608**. In an aspect, the bus circuitry **128** can broadcast data to the usage-based-disturbance circuitry **118** and broadcast other data to the other circuitry **602** at different time periods. To ensure effective communication and reception, the data can be tagged in a manner that allows the intended circuitry to identify whether the data is designated for that circuitry. Implementing the parameter update circuitry **120** using the bus circuitry **128** reduces the amount and complexity of signal routing relative to an architecture in which each of the bank-specific usage-based-disturbance circuits **126** has separate signal routing to the mode register **130**.

[0068] The bus circuitry **128** can transmit data (e.g., broadcast data) through the bus **608** to the bank-specific usage-based-disturbance circuits **126-1** to **126**-N at a first specified time or in a first predetermined mode of operation of the memory device **108**. The bank-specific usage-based-disturbance circuits **126-1** to **126**-N may each receive the data through their respective latches **616**, which are coupled to the bus **608**. The latches **616** enable the bank-specific usage-based-disturbance circuits **126-1** to **126**-N to receive the information that is provided by the bus circuitry **128** and store this information for future reference by the bank-specific usage-based-disturbance circuits **126-1** to **126**-N. a later time during which the bank-specific usage-based-disturbance circuits **126** are activated and able to utilize the information. Thus, a need to transmit the data a plurality of times for the bank-specific usage-based-disturbance circuits **126-1** to **126**-N to each get the data may be obviated.

[0069] The transmission (e.g., broadcast) can encompass various types of information tailored for particular operations. For example, the default parameter **610** can be transmitted during initialization of the memory device **108** to initialize a parameter that is used by the bank-specific usage-based-disturbance circuits **126-1** to **126**-N.

[0070] During a test mode, the test parameter **612** can be transmitted to enable the execution of test procedures and performance evaluations. The bus circuitry **128** broadcasts the test parameter **612** from the auxiliary memory **132** to a test circuitry (which may be the other circuitry **602**) distributed at the local-bank level **122** during the test mode. In the test mode, the memory device **108** may be subjected to various tests to ensure the functioning, reliability, and performance of the memory device **108** meet specifications. During the test mode, different types of tests can be performed, such as tests to check the basic functionality of the memory cells and logic circuits within the memory device **108**, tests to verify the speed or access time of the memory device **108** to ensure it meets the specified performance criteria, tests to evaluate the power consumption and efficiency of the memory device **108**, and tests to subject the memory device **108** to extreme conditions to assess its reliability and stability under adverse situations. Furthermore, other parameters **614** can be transmitted during other operations to the other circuitry **602**.

[0071] In one aspect, the parameter update circuitry **120** broadcast the host-controlled parameter **606** using the bus circuitry **128** based on reception of a mode register write (MRW) command that causes data to be written to the mode register **130**'s operand(s) associated with the host-controlled parameter **606**. The mode register write command can occur during a normal mode of operation of the memory device **108**. The normal mode of operation refers to a time period during which the memory device **108** performs normal read and/or write operations based on commands received from the memory controller **114**. The bus circuitry **128** can complete the broadcast of the host-controlled parameter **606** (or information derived from the host-controlled parameter **606**) to the bank-specific usage-based-disturbance circuits **126-1** to **126**-N during a time period that the banks **410** are idle due to the mode register write command. As such, the parameter update circuitry **120** can ensure that the latest host-controlled parameter **606** is available to the bank-specific usage-based-disturbance circuits **126-1** to **126**-N prior to the end of a row active time (tRAS) and in advance of a next activation (e.g., prior to the banks **410** transitioning from an idle state to an active state).

[0072] The other circuits **604-1** to **604**-Q are each coupled to the bus **608** of the bus circuitry **128**. When the bus circuitry **128** transmits data, at a second specified time or in a second predetermined mode of operation of the memory device **108**, through the bus **608**, the other circuits **604-1** to **604**-

Q may each receive the data. Although not explicitly shown in FIG. **6**, some implementations of the other circuits **604** can include latches **616** to capture and store the information that is broadcasted by the bus circuitry **128**. The broadcasting of different types of information during different time periods and/or during different modes of operation is further described with respect to FIG. **7**.

[0073] FIG. **7** illustrates an example transaction diagram for broadcasting information between the global level **124** and the local-bank level **122** of the memory device **108**. Three different modes of the memory device **108** are depicted in FIG. **7**. These modes include an initialization mode **702**, a normal mode **704**, and a test mode **706**, which can be performed during different time periods. The initialization mode **702** represents a mode of operation during which the memory device **108** initializes itself. The normal mode **704** represents a mode of operation during which the memory device **108** can receive commands from the memory controller **114**. Example commands can include a mode register write command, a read command, and/or a write command. The test mode **706** represents a mode of operation during which the memory device **108** performs a test. Example tests can include a built-in self-test (BIST) or an off-line test. The test can evaluate functionality, reliability, and/or performance of the memory device **108**.

[0074] During the initialization mode **702**, the bus circuitry **128** receives initialization information **708** from the auxiliary memory **132** (Aux. memory **132**) at **710**. The initialization information **708** can include the default parameter **610** and/or the other parameter **614**. At **712**, the bus circuitry **128** broadcasts the default parameter **610** to the usage-based-disturbance circuitry **118**. The usage-based-disturbance circuitry **118** performs an initialization procedure at **714**, which can utilize the default parameter **610**. Optionally at **716**, the bus circuitry **128** broadcasts the other parameter **614** to the other circuitry **602**. The other circuitry **602** can perform an initialization procedure at **718**. Although the broadcasting of the default parameter **610** and the other parameter **614** are shown as separate transactions in FIG. **7**, the bus circuitry **128** can optionally broadcast both the default parameter **610** and the other parameter **614** at the same time. Labels or tags associated with the default parameter **610** and the other parameter **614** can enable the usage-based-disturbance circuitry **118** and the other circuitry **602** to receive the appropriate parameter.

[0075] During the normal mode **704**, the bus circuitry **128** receives the host-controlled parameter **606** from the mode register **130** at **720**. At **722**, the bus circuitry **128** broadcasts the host-controlled parameter **606** (or information derived from the host-controlled parameter **606**) to the usage-based-disturbance circuitry **118**. At **724**, the usage-based-disturbance circuitry performs an operation associated with usage-based-disturbance mitigation. This operation can utilize the host-controlled parameter **606**.

[0076] During the test mode **706**, the bus circuitry **128** receives the test parameter **612** from the auxiliary memory **132** at **726**. At **728**, the bus circuitry **128** broadcasts the test parameter **612** to the other circuitry **602**. At **730**, the other circuitry **602** performs a test based on the test parameter **612**. The operation of the bus circuitry **128** associated with mitigating usage-based disturbance is further described with respect to FIG. **7**.

[0077] FIG. **8** illustrates an example implementation in which the parameter update circuitry **120** can directly pass information from the mode register **130** to the bank-specific usage-based-disturbance circuits **126**. In this example, the parameter update circuitry **120** includes the bus circuitry **128**, the mode register **130**, and the auxiliary memory **132**. The bank-specific usage-based-disturbance circuit **126** includes the latch **616** and array-counter-update (ACU) logic **802** (ACU logic **802**). In an example implementation, the array-counter-update logic **802** can include at least one counter circuit, at least one comparator, and at least one queue.

[0078] The array-counter-update logic **802** performs an array-counter-update procedure, which reads the usage-based-disturbance data **216** and updates the usage-based-disturbance data **216** for an activated row within a corresponding bank **410**. More specifically, the array-counter-update logic **802** updates an activation count **308** associated with an activated row. For instance, during the array-counter-update procedure, the array-counter-update logic **802** reads the activation count **308** that is stored within the activated row, increments the activation count **308** using the counter circuit, and writes the updated activation count to the activated row.

[0079] The array-counter-update logic **802** can also activate mitigation measures based on the usage-based-disturbance data **216** (e.g., based on the activation count **308** or the parity bit **310**). In one instance, the array-counter-update logic **802** compares the activation count **308** to a threshold (e.g., a mitigation threshold) using the comparator circuit and can initiate a refresh operation based on the activation count **308** being larger than the threshold. In this case, the activated row is referred to as an aggressor row. This refresh operation can refresh one or more victim rows that are proximate to (or nearby) the aggressor row.

[0080] In some instances, the memory device **108** may be delayed in performing the refresh operation until sufficient timing resources are available. The array-counter-update logic **802** can continue monitoring the activation count **308** associated with the aggressor row. If the activation count **308** is approaching an intrinsic manufacturer limitation (e.g., an alert threshold), the array-counter-update logic **802** can initiate an alert "back-off" (ABO) procedure. During the alert back-off procedure, the memory device **108** pauses normal operations (e.g., normal read and/or write procedures requested by the host device **104**) for a recovery period during which refresh management (RFM) commands or other functions may be performed in the memory device **108** to mitigate usage-based disturbance. During this recovery period, the victim rows associated with the aggressor row are refreshed.

[0081] Aspects of broadcasting a host-controlled parameter **606** to circuitry distributed at the local-bank level **122** for usage-based-disturbance mitigation enables the host device **104** to control some aspect of the usage-based-disturbance mitigation operation performed by the bank-specific usage-based-disturbance circuit **126**, such as an operation performed by the array-counter-update logic **802**. The specific action or extent of the specific action performed for usage-based disturbance mitigation can be controlled based on information **804** associated with usage-based-disturbance mitigation. The information **804** is provided by the host device **104** (e.g., via the memory controller **114**) and stored by the mode register **130**. By providing an architecture to extemporaneously and dynamically control mitigation behavior of the memory device **108** in a way that is different from a default behavior, the host device **104** can influence and/or adjust how usage-based-disturbance mitigation is performed by the memory device **108**. In example aspects, information associated with the usage-based-disturbance mitigation can be used to update (change a value of) parameters (e.g., thresholds) used during the array-counter update procedure.

[0082] In the illustration of FIG. **8**, an example of the information **804** can be the host-controlled parameter **606** itself. The bus circuitry **128** can broadcast the host-controlled parameter **606** to a plurality of bank-specific usage-based-disturbance circuits **126-1** to **126**-N.

[0083] In an aspect, as the sizes of processors shrink, the impact of usage-based disturbance increases and the intrinsic manufacturer limitation of activation quantities between refreshes decreases. This means that mitigation operations may need to be performed more frequently. The ability to communicate not only default parameters **610** but also host-controlled parameters **606** that are specified by the host device **104** quickly and across multiple bank-specific usage-based-disturbance circuits **126** of the memory device **108** is therefore valuable.

[0084] During an initialization mode **702**, the bus circuitry **128** passes the default parameter **610** from the auxiliary memory **132** to the bank-specific usage-based-disturbance circuit **126**. During the normal mode **704**, the bus circuitry **128** passes the host-controlled parameter **606** to the bank-specific usage-based-disturbance circuit **126**. In this example implementation, the host-controlled parameter **606** represents a mitigation parameter (e.g., mitigation parameter **906** of FIG. **9**), which is used by the array-counter-update logic **802** to perform an operation associated with usage-based-disturbance mitigation. Other implementations of the parameter update circuitry **120** are also possible in which the host-controlled parameter **606** is not directly passed to the bank-specific usage-based-disturbance circuit **126**, as further described with respect to FIG. **9**.

[0085] FIG. **9** illustrates a second example implementation in which the parameter update circuitry **120** can indirectly pass information that is stored in the mode register **130** to the bank-specific usage-based-disturbance circuits **126**. In this example, the parameter update circuitry **120** includes a calculator **902** in addition to the bus circuitry **128**, the mode register **130**, and the auxiliary memory **132**. In example implementations, the calculator **902** can be implemented using a subtractor, a multiplexor, an adder, or other logic circuitry. The calculator **902** processes the information **804** stored within the mode register **130** and provides processed data to the bus circuitry **128** for broadcasting to the bank-specific usage-based-disturbance circuits **126**. In this manner, the host-controlled parameter **606** that is stored within the mode register **130** is not directly passed to the bank-specific usage-based-disturbance circuit **126** and is instead used to generate other data (e.g., a mitigation parameter **906**) that is broadcasted to the bank-specific usage-based-disturbance circuits **126**.

[0086] In an aspect, the mode register **130** receives information **804** from the memory controller **114**. This information **804** represents an adjustment parameter **904**, which is the host-controlled parameter **606**. The bank-specific usage-based-disturbance circuit **126** is unable to directly use the adjustment parameter **904**. However, the calculator **902** uses the adjustment parameter **904** to determine (or generate) a mitigation parameter **906**, which can be used by the bank-specific usage-based-disturbance circuit **126**. In some examples, the calculator **902** generates the mitigation parameter **906** based on the adjustment parameter **904** and the default parameter **610**. For instance, the calculator **902** uses the adjustment parameter **904** to adjust the default parameter **610**.

[0087] In an example implementation, the mitigation parameter **906** can be a mitigation threshold that is used by the usage-based-disturbance circuitry **118** for mitigation operations. In particular, the array-counter-update logic **802** can compare the activation count **308** of an activated row to the mitigation parameter **906** to determine whether or not to initiate a refresh operation to mitigate usage-based disturbance. The mitigation threshold is typically set as a value that is significantly smaller than the maximum number of times a row **302** can be safely activated before a usage-based-disturbance mitigation operation becomes necessary. In this case, the default parameter **610** can represent a default threshold **908**, and the mitigation parameter **906** represents an updated value of the default threshold **908** (e.g., an adaptive mitigation threshold).

[0088] In another example implementation, the mitigation parameter **906** can be an alert threshold that is used by the usage-based-disturbance circuitry **118** to trigger an alert back-off mode. During the alert back-off mode, a denial-of-service situation can occur during which the memory device **108** dedicates resources to refreshing victim rows to mitigate usage-based disturbance. The alert threshold is typically set as a value that is smaller than the maximum number of times a row **302** can be safely activated as determined by the manufacturer of the memory device **108**. The alert threshold is also typically higher than the mitigation threshold. The calculator **902** can determine a value of the alert threshold by modifying a value of the default parameter **610** in accordance with the adjustment parameter **904** stored by the mode register **130**.

[0089] During a first mode of the memory device **108** (e.g., during an initialization mode **702**), the bus circuitry **128** selectively passes (e.g., broadcasts) the default parameter **610** from the auxiliary memory **132** to the usage-based-disturbance circuitry **118** distributed at the local-bank level **122**. During a second mode of the memory device **108** (e.g., during a normal operational mode **704**), the bus circuitry **128** passes the mitigation parameter **906** from the mode register **130** to the usage-based-disturbance circuitry **118**. In this sense, the bus circuitry **128** can pass the default parameter **610** and the mitigation parameter **906** during different time intervals. For example, the default parameter **610** can be passed to the usage-based-disturbance circuitry **118** at initialization of the memory device **108**, and the mitigation parameter **906** can be passed to the usage-based-disturbance circuitry **118** during normal operation of the memory device **108**. Of course, these are not meant to be limiting, as other modes or operations and time intervals can be obtained in view of the descriptions herein.

[0090] In some implementations of the memory device **108**, the parameter update circuitry **120** passes both the host-controlled parameter **606** and the mitigation parameter **906**. In general, the parameter update circuitry **120** can be designed to broadcast any combination of one or more host-controlled parameters **606** and one or more mitigation parameters **906**. In a first example, the parameter update circuitry **120** broadcasts one host-controlled parameter **606** and/or one mitigation parameter **906**. In a second example, the parameter update circuitry **120** broadcasts multiple host-controlled parameters **606** and/or multiple mitigation parameters **906**.
[0091] As shown in FIGS. **8** and **9**, the bank-specific usage-based-disturbance circuit **126** can include a corresponding latch **616**, which is adapted to store the broadcasted parameter (or parameters). The array-counter-update logic **802** is then provided with the stored parameter from the latch **616** when appropriate. For example, in an aspect, the array-counter-update logic **802** performs the array update procedure based on a pre-charge command. The broadcasting of the host-controlled parameter **606** and/or the mitigation parameter **906** (e.g., information derived from the host-controlled parameter **606**) may thus occur prior to the pre-charge command. More specifically, the broadcasting may be based on (or occur responsive to) reception of a mode register write command that provides the information **804** for storage within the mode register **130**. The mode register write command can optionally change or adjust the information **804** stored within the mode register **130**.
[0092] The mode register write command is associated with a time period during which the banks **410** are in an idle mode. The parameter update circuitry **120** is able to complete the broadcasting of the host-controlled parameter **606** or the mitigation parameter **906** within the time period during which the banks **410** are in the idle mode. In effect, there is sufficient time to transmit the host-controlled parameter **606** and/or the mitigation parameter **906** to all of the bank-specific usage-based-disturbance circuits **126** before the ACU logic **802** utilizes the host-controlled parameter **606** or the mitigation parameter **906** to perform the array counter update procedure. The broadcasting can also occur during the normal mode **704**. In some cases, the parameter update circuitry **120** includes additional logic that can determine that information stored in the mode register **130** has changed and activates the bus circuitry **128** to pass the information to the usage-based-disturbance circuitry **118**.
Example Method
[0093] This section describes example methods for broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation with reference to the flow diagrams of FIGS. **10** and **11**. These descriptions may also refer to components, entities, and other aspects depicted in FIG. **1** to FIG. **9** by way of example only. The described methods are not necessarily limited to performance by one entity or multiple entities operating on one device.
[0094] FIG. **10** illustrates a flow diagram **1000**, which includes operations **1002** through **1008**. In aspects, operations of the method **1000** are implemented by or with parameter update circuitry **120** as described with reference to FIG. **1** to FIG. **9**. At **1002**, information provided by a memory controller is stored in at least one mode register of a memory device. The information is associated with a mitigation parameter that is used in an operation that mitigates usage-based disturbance within the memory device. For example, the mode register **130** of the memory device **108** stores information **804**, which represents a host-controlled parameter **606**, as shown in FIGS. **6**, **8** and **9**. The information **804** is associated with the mitigation parameter **906** that is used in an operation that mitigates usage-based disturbance within the memory device **108**. For example, the information **804** can include the mitigation parameter **906** itself (as shown in FIG. **8**) or information that is used to derive the mitigation parameter **906** (as shown in FIG. **9**). The mitigation parameter **906** is used by the bank-specific usage-based-disturbance circuits **126** (e.g., the array-counter-update logic **802**) to mitigate usage-based disturbance within corresponding banks **410**.
[0095] At **1004**, a default parameter is stored in an auxiliary memory of the memory device. The

default parameter represents a default value of the mitigation parameter. For example, the auxiliary memory **132** stores the default parameter **610**, as shown in FIG. **6**. In an example implementation, the auxiliary memory **132** is non-volatile memory, such as an array of fuses and logic for reading information stored in the fuses.

[0096] At **1006**, the default parameter is broadcasted, using bus circuitry of the memory device, to usage-based-disturbance circuitry that is distributed at a local-bank level of the memory device. The broadcasting of the default parameter occurs based on the memory device operating in accordance with a first mode. For example, the bus circuitry **128** broadcasts the default parameter **610** to the usage-based-disturbance circuitry **118** that is distributed at the local-bank level **122** of the memory device **108**, as shown in FIG. **7**, **8**, or **9**. The broadcasting of the default parameter **610** occurs based on the memory device operating in accordance with a first mode, such as the initialization mode **702** of FIG. **7**.

[0097] At **1008**, the information is broadcasted, using the bus circuitry, to the usage-based-disturbance circuitry. The broadcasting of the information occurs based on the memory device operating in accordance with a second mode. For example, the bus circuitry **128** broadcasts the information **804** (e.g., the host-controlled parameter **606** or information derived from the host-controlled parameter **606** such as the mitigation parameter **906**) to the usage-based-disturbance circuitry **118**, as shown in FIG. **7**, **8**, or **9**. The broadcasting of the information **804** occurs based on the memory device **108** operating in accordance with a second mode, such as the normal mode **704** of FIG. **7**. In some cases, the broadcasting is triggered by a mode register write operation. As described above, the bus circuitry **128** can broadcast different information to the usage-based-disturbance circuitry **118** based on different modes of the memory device **108**.

[0098] FIG. **11** illustrates a flow diagram **1100**, which includes operations **1102** through **1108**. In aspects, operations of the method **1100** are implemented by or with parameter update circuitry **120** as described with reference to FIG. **1** to FIG. **9**. At **1102**, information associated with mitigating usage-based disturbance within the memory device is stored in a mode register of the memory device. For example, the mode register **130** of the memory device **108** stores information **804** associated with mitigating usage-based disturbance, as shown in FIGS. **6**, **8**, and **9**. This information **804** represents at least one host-controlled parameter **606**.

[0099] At **1104**, other information associated with a test mode is stored in an auxiliary memory of the memory device. For example, the auxiliary memory **132** of the memory device **108** stores a test parameter **612**, as shown in FIG. **6**.

[0100] At **1106**, the information associated with mitigating the usage-based disturbance is broadcasted, during a first time period, to usage-based-disturbance circuitry that is distributed at a local-bank level. For example, the bus circuitry **128** broadcasts the information **804** to the usage-based-disturbance circuitry **118**, which is distributed at the local-bank level **122**, as shown in FIG. **6**. This broadcasting occurs during a first time period, as shown in FIG. **7**.

[0101] At **1108**, the other information associated with the test mode is broadcasted, during a second time period, to other circuitry that is distributed at the local-bank level. For example, the bus circuitry **128** broadcasts the other information (e.g., the test parameter **612**) to the other circuitry **602**, which is distributed at the local-bank level **122**, as shown in FIG. **6**. This broadcasting occurs during a second time period, as shown in FIG. **8**. The second time period is different than (e.g., does not overlap) the first time period.

[0102] For the figures described above, the orders in which operations are shown and/or described are not intended to be construed as a limitation. Any number or combination of the described process operations can be combined or rearranged in any order to implement a given method or an alternative method. Operations may also be omitted from or added to the described methods. Further, described operations can be implemented in fully or partially overlapping manners.

[0103] Aspects of these methods may be implemented in, for example, hardware (e.g., fixed-logic circuitry or a processor in conjunction with a memory), firmware, software, or some combination thereof. The methods may be realized using one or more of the apparatuses or components shown in FIG. **1** to FIG. **8**, the components of which may be further divided, combined, rearranged, and so on. The devices and components of these figures generally represent hardware, such as electronic devices, packaged modules, IC chips, or circuits; firmware or the actions thereof; software; or a combination thereof. Thus, these figures illustrate some of the many possible systems or apparatuses capable of implementing the described methods.

[0104] Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program (e.g., an application) or data from one entity to another. Non-transitory computer storage media can be any available medium accessible by a computer, such as RAM, ROM, Flash, EEPROM, optical media, and magnetic media.

[0105] In the following, various examples for implementing aspect of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation are described:

[0106] Example 1: An apparatus comprising: [0107] a memory device comprising: [0108] usage-based-disturbance circuitry distributed at a local-bank level of the memory device, the usage-based-disturbance circuitry configured to perform an operation that mitigates usage-based disturbance within the memory device; [0109] at least one mode register configured to store information provided by a memory controller that is coupled to the memory device, the information being associated with a mitigation parameter that is utilized by the usage-based-disturbance circuitry to perform the operation that mitigates the usage-based disturbance; [0110] an auxiliary memory configured to store a default parameter, the default parameter representing a default value of the mitigation parameter; and [0111] bus circuitry implemented at a global level and coupled to the usage-based-disturbance circuitry, the at least one mode register, and the auxiliary memory, the bus circuitry configured to selectively: [0112] pass the default parameter from the auxiliary memory to the usage-based-disturbance circuitry in accordance with a first mode of the memory device; and [0113] pass the information stored in the at least one mode register to the usage-based-disturbance circuitry in accordance with a second mode of the memory device.

[0114] Example 2: The apparatus of example 1 or any other example, wherein: [0115] the memory device further comprises a memory array; and [0116] the usage-based-disturbance circuitry is configured to perform the operation that mitigates the usage-based disturbance within the memory array based on the information provided by the at least one mode register or the default parameter.

[0117] Example 3: The apparatus of example 2 or any other example, wherein: [0118] the information stored by the at least one mode register comprises a mitigation threshold; and [0119] the usage-based-disturbance circuitry is configured to: [0120] compare an activation count stored in an activated row of the memory array with the mitigation threshold; and [0121] perform the operation based on the activation count exceeding the mitigation threshold.

[0122] Example 4: The apparatus of example 2 or any other example, wherein: [0123] the information stored by the at least one mode register comprises an alert threshold; and [0124] the usage-based-disturbance circuitry is configured to: [0125] compare an activation count stored in an activated row of the memory array with the alert threshold; and [0126] trigger the memory device to execute an alert back-off procedure responsive to the activation count exceeding the alert threshold.

[0127] Example 5: The apparatus of example 1 or any other example, wherein the usage-based-disturbance circuitry comprises a plurality of bank-specific usage-based-disturbance circuits that are coupled to different banks of the memory device and configured to mitigate the usage-based disturbance for the different banks.

[0128] Example 6: The apparatus of example 5 or any other example, wherein: [0129] each bank-specific usage-based-disturbance circuit of the plurality of bank-specific usage-based-disturbance circuits comprises a latch configured to store information from the bus circuitry; and [0130] the bus circuitry comprises a communication bus coupled to the latch for each of the bank-specific usage-based-disturbance circuits.

[0131] Example 7: The apparatus of example 1 or any other example, wherein: [0132] the information provided by the memory controller comprises an adjustment parameter; and [0133] the apparatus further comprises: [0134] a calculator configured to determine a value of the mitigation parameter by modifying a value of the default parameter in accordance with the adjustment parameter.

[0135] Example 8: The apparatus of example 7 or any other example, wherein the calculator comprises a subtractor or a multiplexer.

[0136] Example 9: The apparatus of example 1 or any other example, wherein the auxiliary memory comprises a fuse array or flash memory.

[0137] Example 10: The apparatus of example 1 or any other example, wherein: [0138] the bus circuitry is configured to pass the information from the at least one mode register to the usage-based-disturbance circuitry based on a mode-register write command associated with the at least one mode register; and [0139] the bus circuitry is configured to pass the information from the at least one mode register to the usage-based-disturbance circuitry within a time period during which banks of the memory device are idle based on the mode-register write command.

[0140] Example 11: The apparatus of example 1 or any other example, wherein: [0141] the memory device comprises other circuitry distributed at the local-bank level, the other circuitry configured to perform an operation associated with a test mode; [0142] the auxiliary memory is configured to store information that is utilized by the other circuitry to perform the operation associated with the test mode; and [0143] the bus circuitry is configured to pass the information that is utilized by the other circuitry from the auxiliary memory to the other circuitry.

[0144] Example 12: A method comprising: [0145] storing, in at least one mode register of a memory device, information provided by a memory controller, the information associated with a mitigation parameter that is used in an operation that mitigates usage-based disturbance within the memory device; [0146] storing, in an auxiliary memory of the memory device, a default parameter representing a default value of the mitigation parameter; [0147] broadcasting, using bus circuitry of the memory device, the default parameter to usage-based-disturbance circuitry that is distributed at a local-bank level of the memory device, the broadcasting of the default parameter occurring based on the memory device operating in accordance with a first mode; and [0148] broadcasting, using the bus circuitry, the information to the usage-based-disturbance circuitry, the broadcasting of the information occurring based on the memory device operating in accordance with a second mode.

[0149] Example 13: The method of example 12 or any other example, wherein: [0150] the first mode comprises an initialization mode of the memory device; and [0151] the second mode comprises a normal operational mode of the memory device.

[0152] Example 14: The method of example 12 or any other example, wherein the broadcasting of the information further comprises broadcasting the information responsive to a change in the information stored in the at least one mode register.

[0153] Example 15: The method of example 14 or any other example, wherein the broadcasting of the information further comprises: [0154] receiving a mode-register write command that changes the information stored in the at least one mode register and causes banks of the memory device to become idle for a time period; and [0155] completing the broadcasting of the information to the usage-based-disturbance circuitry within the time period during which the banks are idle.

[0156] Example 16: The method of example 12 or any other example, wherein: [0157] the information provided by the memory controller comprises an adjustment parameter; [0158] the broadcasting of the information to the usage-based-disturbance circuitry comprises: [0159] computing, by a calculator of the memory device, an updated value of the mitigation parameter using the adjustment parameter and the default parameter; and [0160] broadcasting the updated value of the mitigation parameter to the usage-based-disturbance circuitry.

[0161] Example 17: The method of example 12 or any other example, further comprising: [0162] storing, in the auxiliary memory, other information associated with a test mode of the memory device; and [0163] broadcasting, using the bus circuitry, the other information associated with the test mode to other circuitry distributed at the local-bank level, the broadcasting of the other information occurring based on the memory device operating in accordance with the test mode.
[0164] Example 18: An apparatus comprising: [0165] a memory device comprising: [0166] at least one mode register configured to store information associated with mitigating usage-based disturbance within the memory device; [0167] an auxiliary memory configured to store other information associated with a test mode; [0168] bus circuitry coupled to the at least one mode register and the auxiliary memory, the bus circuitry configured to selectively: [0169] broadcast the information associated with mitigating the usage-based disturbance to usage-based-disturbance circuitry distributed at a local-bank level during a first time period; and [0170] broadcast the other information associated with the test mode to other circuitry distributed at the local-bank level during a second time period.
[0171] Example 19: The apparatus of example 18 or any other example, wherein: [0172] the memory device is configured to selectively: [0173] operate in accordance with a normal operational mode during the first time period; and [0174] operate in accordance with a test mode during the second time period.
[0175] Example 20: The apparatus of example 19 or any other example, wherein: [0176] the auxiliary memory is configured to store a default parameter that represents a default value of the information; [0177] the bus circuitry is configured to selectively broadcast the default parameter to the usage-based-disturbance circuitry during a third time period; and [0178] the memory device is configured to selectively operate in accordance with an initialization mode during the third time period.
[0179] Unless context dictates otherwise, use herein of the word “or” may be considered use of an “inclusive or,” or a term that permits inclusion or application of one or more items that are linked by the word “or” (e.g., a phrase “A or B” may be interpreted as permitting just “A,” as permitting just “B,” or as permitting both “A” and “B”). Also, as used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. For instance, “at least one of a, b, or c” can cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiples of the same element (e.g., a-a, a-a-a, a-a-b, a-a-c, a-b-b, a-c-c, b-b, b-b-b, b-b-c, c-c, and c-c-c, or any other ordering of a, b, and c). Further, items represented in the accompanying figures and terms discussed herein may be indicative of one or more items or terms, and thus reference may be made interchangeably to single or plural forms of the items and terms in this written description.
CONCLUSION
[0180] Although aspects of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation have been described in language specific to certain features and/or methods, the subject of the appended claims is not necessarily limited to the specific features or methods described. Rather, the specific features and methods are disclosed as a variety of example implementations of broadcasting a host-controlled parameter to circuitry distributed at a local-bank level for usage-based-disturbance mitigation.

Claims

1. An apparatus comprising: a memory device comprising: usage-based-disturbance circuitry distributed at a local-bank level of the memory device, the usage-based-disturbance circuitry configured to perform an operation that mitigates usage-based disturbance within the memory device; at least one mode register configured to store information provided by a memory controller that is coupled to the memory device, the information being associated with a mitigation parameter

that is utilized by the usage-based-disturbance circuitry to perform the operation that mitigates the usage-based disturbance; an auxiliary memory configured to store a default parameter, the default parameter representing a default value of the mitigation parameter; and bus circuitry implemented at a global level and coupled to the usage-based-disturbance circuitry, the at least one mode register, and the auxiliary memory, the bus circuitry configured to selectively: pass the default parameter from the auxiliary memory to the usage-based-disturbance circuitry in accordance with a first mode of the memory device; and pass the information stored in the at least one mode register to the usage-based-disturbance circuitry in accordance with a second mode of the memory device.
2. The apparatus of claim 1, wherein: the memory device further comprises a memory array; and the usage-based-disturbance circuitry is configured to perform the operation that mitigates the usage-based disturbance within the memory array based on the information provided by the at least one mode register or the default parameter.
3. The apparatus of claim 2, wherein: the information stored by the at least one mode register comprises a mitigation threshold; and the usage-based-disturbance circuitry is configured to: compare an activation count stored in an activated row of the memory array with the mitigation threshold; and perform the operation based on the activation count exceeding the mitigation threshold.
4. The apparatus of claim 2, wherein: the information stored by the at least one mode register comprises an alert threshold; and the usage-based-disturbance circuitry is configured to: compare an activation count stored in an activated row of the memory array with the alert threshold; and trigger the memory device to execute an alert back-off procedure responsive to the activation count exceeding the alert threshold.
5. The apparatus of claim 1, wherein the usage-based-disturbance circuitry comprises a plurality of bank-specific usage-based-disturbance circuits that are coupled to different banks of the memory device and configured to mitigate the usage-based disturbance for the different banks.
6. The apparatus of claim 5, wherein: each bank-specific usage-based-disturbance circuit of the plurality of bank-specific usage-based-disturbance circuits comprises a latch configured to store information from the bus circuitry; and the bus circuitry comprises a communication bus coupled to the latch for each of the bank-specific usage-based-disturbance circuits.
7. The apparatus of claim 1, wherein: the information provided by the memory controller comprises an adjustment parameter; and the apparatus further comprises: a calculator configured to determine a value of the mitigation parameter by modifying a value of the default parameter in accordance with the adjustment parameter.
8. The apparatus of claim 7, wherein the calculator comprises a subtractor or a multiplexer.
9. The apparatus of claim 1, wherein the auxiliary memory comprises a fuse array or flash memory.
10. The apparatus of claim 1, wherein: the bus circuitry is configured to pass the information from the at least one mode register to the usage-based-disturbance circuitry based on a mode-register write command associated with the at least one mode register; and the bus circuitry is configured to pass the information from the at least one mode register to the usage-based-disturbance circuitry within a time period during which banks of the memory device are idle based on the mode-register write command.
11. The apparatus of claim 1, wherein: the memory device comprises other circuitry distributed at the local-bank level, the other circuitry configured to perform an operation associated with a test mode; the auxiliary memory is configured to store information that is utilized by the other circuitry to perform the operation associated with the test mode; and the bus circuitry is configured to pass the information that is utilized by the other circuitry from the auxiliary memory to the other circuitry.
12. A method comprising: storing, in at least one mode register of a memory device, information provided by a memory controller, the information associated with a mitigation parameter that is used in an operation that mitigates usage-based disturbance within the memory device; storing, in

an auxiliary memory of the memory device, a default parameter representing a default value of the mitigation parameter; broadcasting, using bus circuitry of the memory device, the default parameter to usage-based-disturbance circuitry that is distributed at a local-bank level of the memory device, the broadcasting of the default parameter occurring based on the memory device operating in accordance with a first mode; and broadcasting, using the bus circuitry, the information to the usage-based-disturbance circuitry, the broadcasting of the information occurring based on the memory device operating in accordance with a second mode.
13. The method of claim 12, wherein: the first mode comprises an initialization mode of the memory device; and the second mode comprises a normal operational mode of the memory device.
14. The method of claim 12, wherein the broadcasting of the information further comprises broadcasting the information responsive to a change in the information stored in the at least one mode register.
15. The method of claim 14, wherein the broadcasting of the information further comprises: receiving a mode-register write command that changes the information stored in the at least one mode register and causes banks of the memory device to become idle for a time period; and completing the broadcasting of the information to the usage-based-disturbance circuitry within the time period during which the banks are idle.
16. The method of claim 12, wherein: the information provided by the memory controller comprises an adjustment parameter; the broadcasting of the information to the usage-based-disturbance circuitry comprises: computing, by a calculator of the memory device, an updated value of the mitigation parameter using the adjustment parameter and the default parameter; and broadcasting the updated value of the mitigation parameter to the usage-based-disturbance circuitry.
17. The method of claim 12, further comprising: storing, in the auxiliary memory, other information associated with a test mode of the memory device; and broadcasting, using the bus circuitry, the other information associated with the test mode to other circuitry distributed at the local-bank level, the broadcasting of the other information occurring based on the memory device operating in accordance with the test mode.
18. An apparatus comprising: a memory device comprising: at least one mode register configured to store information associated with mitigating usage-based disturbance within the memory device; an auxiliary memory configured to store other information associated with a test mode; bus circuitry coupled to the at least one mode register and the auxiliary memory, the bus circuitry configured to selectively: broadcast the information associated with mitigating the usage-based disturbance to usage-based-disturbance circuitry distributed at a local-bank level during a first time period; and broadcast the other information associated with the test mode to other circuitry distributed at the local-bank level during a second time period.
19. The apparatus of claim 18, wherein: the memory device is configured to selectively: operate in accordance with a normal operational mode during the first time period; and operate in accordance with a test mode during the second time period.
20. The apparatus of claim 19, wherein: the auxiliary memory is configured to store a default parameter that represents a default value of the information; the bus circuitry is configured to selectively broadcast the default parameter to the usage-based-disturbance circuitry during a third time period; and the memory device is configured to selectively operate in accordance with an initialization mode during the third time period.